

Software Manual

Version 1.0.1

SENSO-Stick



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Revisions

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23.03.2015	1.0.0	Software Manual SENSO-Stick
17.03.2016	1.0.1	Added English translation, contact data updated, several smaller changes

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1 Introduction

1.1 About this Software Manual

- Prerequisite for safe handling and failure-free use of the SENSO-Stick is the knowledge of the basic safety rules and regulations.
- This Software Manual contains the most important information on how to use the SENSO-Stick safely.
- All personnel using the SENSO-Stick are to observe this Software Manual as well as the Operating Manual, particularly the safety rules and regulations.
- Moreover the rules and regulations in effect at the specific place of operation are to be observed.
- This manual is currently available in:
 - German (original manual)
 - English (translation of the original manual)

1.2 Designated Use

The SENSO-Stick is a 3-axis input device with integrated drives for generating the reacting forces (force feedback joystick). It is exclusively designed for using in simulation, training, research and development. Any other use or any use beyond that is misuse. Using the device in steering systems for vehicles employed in regular road traffic is strictly forbidden. SENSODRIVE GmbH cannot be held liable for any damages resulting of such usage.

Designated use includes paying regard to all information in this operating manual.

2 Technical Data

2.1 Communication Interface

Communication with the SENSO-Stick by
CAN.

Preset baud rate of
1 Mbit/s = 1 Mbaud.

2.2 Definition of the Axes

Figure 2-1 shows the top view of the SENSO-Stick. Located in this are the directions of movement of the handle.

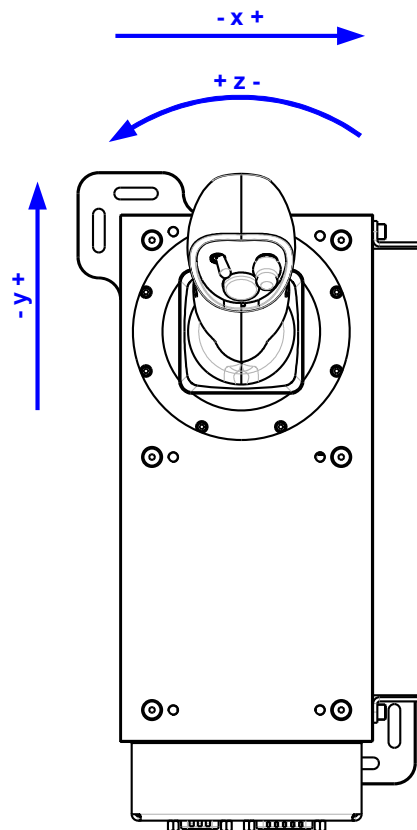


Fig. 2-1: Definition of the movement direction

An upward movement of the handle corresponds to a positive movement in the y-direction. If you move the handle to the right, this represents a positive movement in the x-direction. Upon rotation of the handle counterclockwise one receives the positive angle of rotation z.



NOTICE

A positive motor torque rotates the SENSO-Stick in a positive direction.

3 Safety Instructions

3.1 Designated Use

Please refer to chapter 1.2 for information on the designated use of the SENSO-Stick.

3.2 Instructions on Operation and Use

Operator, in the scope of this document, is anyone who integrates the SENSO-Stick in a simulator and/or operates in conjunction with a host controller (host). In contrast, the user is the one who directs the SENSO-Stick. The operator is responsible for the proper use of the SENSO-Stick according to the directions stated in Chapter 3.2.2.

Non-observance of following instructions can cause serious injury.

	WARNING Automatic Moving Parts The SENSO-Stick can move automatically when instructed by the host controller
	WARNING There is a potential risk of hand and arm injuries (e.g. in the area of the stopper) Keep hands and arms away

3.2.1 Instructions on the Operation of the SENSO-Stick

- The SENSO-Stick is not an apparatus in the sense of the “Law on Equipment Safety”, the “Law on EMC” or the “EG Machinery Directive”. It is a component. It is only to be used in conjunction with a host. The SENSO-Stick may not be put into operation until the whole apparatus that the SENSO-Stick has to be built into, has been put together and is in accordance with the European Directives.
- It is the responsibility of the operator to ensure that the combination of the SENSO-Stick together with the host complies with the national safety regulations.
- Do not use the SENSO-Stick without monitoring the communication between the host and the SENSO-Stick. The host should monitor the communication at all times (Host Watchdog).
- On no accounts should the safety functions of the SENSO-Stick be de-activated.
- It is the responsibility of the operator to make sure that the SENSO-Stick behaves like expected.
- Limit the torque to 20% while the SENSO-Stick is put in operation (Compare Chapter 5.4 and 6.2.2).

3.2.2 Instructions on the Use of the SENSO-Stick

- Inexperienced users must be supervised while using the SENSO-Stick.
- It is forbidden to put fingers, hands or any other body parts into the stop area of the joystick handle. The stopper area is located below the handle between the plastic stopper and the metal shaft of the handle. It limits the range of movement of the joystick handle.
- Children below the age of fourteen are not allowed to use the SENSO-Stick.

3.2.3 Monitoring functions of the SENSO-Stick

- The communication with the host controller is monitored by the CAN Watchdog (Chapter 6.2.2.3).
- The Alive Counter monitors the loss of the Normal Mode messages during the communication with the host controller (Chapter 6.2.2.3).
- The state machine prevents an automatic restart after an error.
- The CPU Watchdog monitors the run time behavior of the SENSO-Stick firmware.
- The enable button checks whether the joystick handle is being held.

4 System Overview of the SENSO-Stick

The SENSO-Stick consists of three brushless DC motors (SENSO-Stick Motor) each having a control device (SENSO-Stick Controller). Fig. 4-2 shows schematically the construction of modular and the integrated version of the SENSO-Stick.

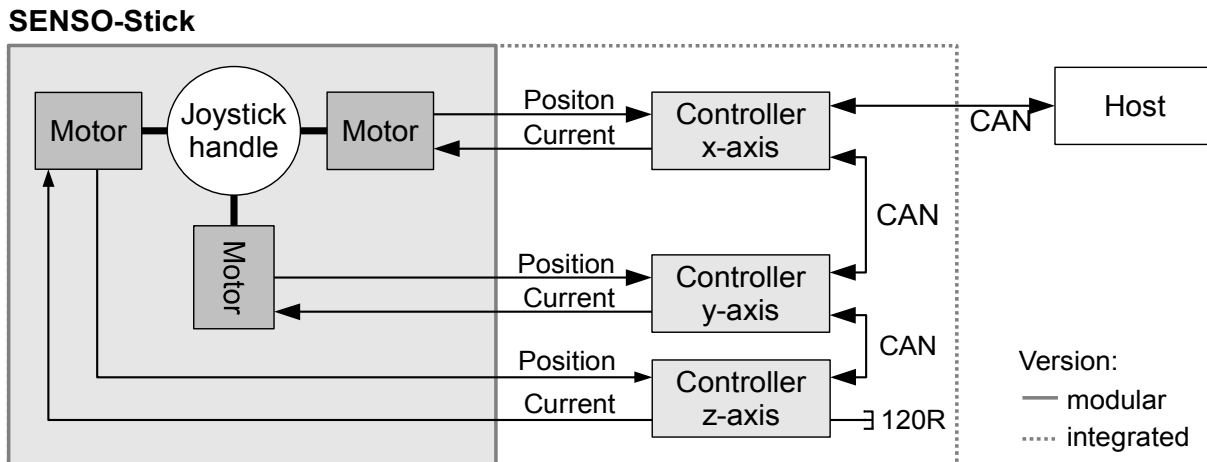


Fig. 4-1: System overview of the SENSO-Stick

The SENSO-Stick controller takes over the control of the motor torque as well as the communication between the SENSO-Stick axis and the host controller. The communication runs via a 1Mbit/s CAN bus connection. For a short introduction into CAN communication, please refer to Chapter 5.

The control firmware consists of a control module for general tasks and three modules for the three other operations (Fig. 4-2). These are: Normal Mode, Basic Mode and Referencing Mode.

Independent control tasks e.g. CAN Watchdog, end stops or overload protection of the motor, are taken care of by the control module. It also activates and deactivates the different operating modes based on a control word from the host.

The normal mode is the standard operation of the controller firmware. Aside from commanding torque this mode offers three more parameters to create a realistic feeling when operating: friction, damping and spring stiffness. The SENSO-Stick returns torque, position and velocity back to the host. Mechanical end stops can be simulated. The cycle time is 1ms.

In case a normal mode cycle time is too slow, the basic mode can be selected. This mode is interrupt-driven. Every incoming CAN message triggers an interrupt and provides the desired torque straight to the current controller. To avoid overload on the CAN bus the features are limited. Only desired torque and actual position of the steering wheel are transmitted. This operation is only advantageous when the cycle time can be realized at smaller than 1ms. The minimal cycle time is 300µs.

The SENSO-Stick does not have an absolute position signal. After power-up the position is independent of the actual zero position. To move the joystick after a new start in the zero position the Referencing Mode has to be used. Chapter 6.5 shows this in detail

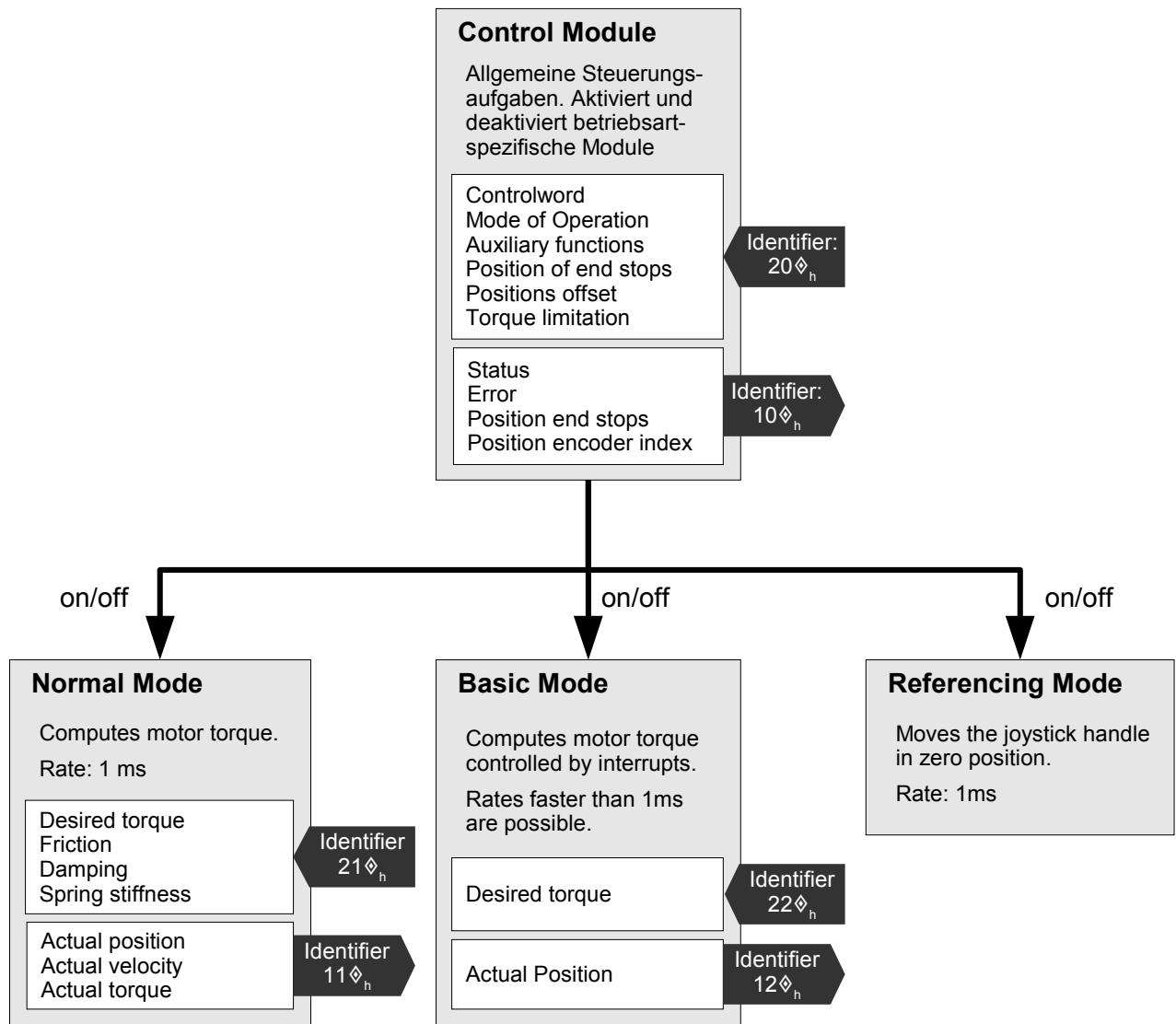


Fig. 4-2: Firmware construction of the SENSO-Stick control firmware. The last digit of the identifier indicates the axis. It is characterized as " $\diamond$ ".

5 Quickstart: Communication with the SENSO-Stick

The communication with the SENSO-Stick and its functions are described in detail in Chapter 6. This chapter is meant to be a quickstart guide and offers a brief first introduction into CAN field bus.

5.1 General introduction to CAN

A CAN message contains following information:

- A “CAN-Identifier” to identify the message
- A “DLC” (data Length Code) to indicate the number of data bytes
- 0 to 8 data bytes.

<i>Identifier</i>	<i>DLC</i>	<i>Data</i>							
		<i>Byte 1</i>	<i>Byte 2</i>	<i>Byte 3</i>	<i>Byte 4</i>	<i>Byte 5</i>	<i>Byte 6</i>	<i>Byte 7</i>	<i>Byte 8</i>
201 _h	8	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

The SENSO-Stick uses identifiers in standard format (11 bit) in the range 101_h ... 2B3_h. The used baud rate is 1Mbits/s (1 Mbaud).

Please take note that 16- or 32-bit values are noted with the least significant byte first. This is depicted as “Little Endian”. The number 1234_h in the first bytes is thus written this way:

$$4660 = 1234_{\text{h}} =$$

<i>Data</i>	
<i>Byte 1</i>	<i>Byte 2</i>
34 _h	12 _h

5.2 CAN Kommunikation with the SENSO-Stick

5.2.1 Bitwise Composition of the CAN Identifier

Figure 5-1 shows the logical construction of the CAN Identifier (CAN-ID)

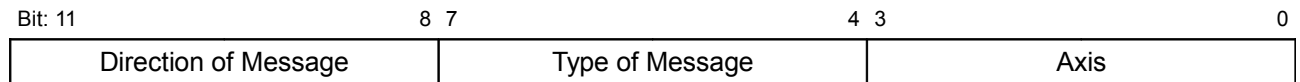


Fig. 5-1: Construction of the CAN-Identifier

The four least significant bits (bits 0-3) of the identifier determine which axes are addressed. The four bits with higher order (bits 4-7) contain the type of message. The direction of the message is cleared by the most significant bits (bits 8-11).

5.2.2 Breakdown of the CAN-IDs

The following tables illustrate the breakdown of CAN identifiers

<i>Bit</i>	<i>Description</i>
0-3	1_h x-Axis 2_h y-Axis 3_h z-Axis
4-7	0_h ... 9_h Standard Message A_h ... F_h Parameter Message <i>Key: see following table</i>
8-11	0_h Firmware Update 1_h Message to SENSO-Stick 2_h Message to Host (Return Values)

5.3 Step 1: Test of Communication

In the following tables CAN messages sent from the host to the SENSO-Stick are highlighted gray. White backgrounds indicate responses sent from the SENSO-Stick to the host. In the identifier field is the last digit of the identifier for the axis of SENSO-Stick. “ $\diamond$ ” represents a placeholder for the respective axis of the SENSO-Stick.

- 1) To check the communication send the following CAN message to the control module (Identifier 20 $\diamond$ _h: Normal Mode x-, y- respectively z-axis, motor switched off; see Chapter 6.2.2 and 6.2.3):

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond$ _h	8	10 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

The SENSO-Stick responds with an acknowledge message with identifier 10 $\diamond$ _h:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond$ _h	8	Status	Status	Error	Error	Endstop	Endstop	Enc.-Index	Enc.-Index

For a more detailed description of the Status and error messages please refer to Chapter 6.2.3.

- 2) To check motor position encoder send a CAN message in Normal Mode (Identifier 21 $\diamond$ _h; compare Chapter 6.3):

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
21 $\diamond$ _h	8	00 _h	00 _h	00 _h	00 _h	00 _h	01 _h	00 _h	00 _h

Byte 1-2: desired torque
 Byte 3-4: friction
 Byte 5: damping
 Byte 6: Alive-Counter
 Byte 7-8: spring stiffness

The SENSO-Stick responds with an acknowledging message with identifier 11 $\diamond$ _h:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
11 $\diamond$ _h	8	Position	Position	Position	Alive-Counter	Velocity	Velocity	Torque	Torque

Byte 1-3: Position of the handle in increments
 Byte 4: Alive-Counter
 Byte 5-6: Motor speed in rpm

While the joystick handle is moved the motor encoder may be checked several times by resending message 21 $\diamond$ _h. Example for the calculation of the position

<i>Position</i>		
Byte 1	Byte 2	Byte 3
B3 _h	01 _h	00 _h

These CAN data result in a Position = 1B3_h [inc]= 435 [inc].

This corresponds, with an assumed encoder resolution of 24000 inc/revolution, to an angle of 6.53°.

5.4 Step 2: Referencing of the SENSO-Stick

As soon as the motor has been turned on, CAN messages have to be sent periodically (T < 50ms). Otherwise the CAN Watchdog of the SENSO-Stick will generate an error and the motor will shut down (refer to Chapter 6.2.2.3). This applies to any CAN message and each axis e.g. the CAN Watchdog is reset regardless of any CAN identifier.

- 1) If the error read out from the SENSO-Stick was 0000_h, you can move onto point 2. Otherwise this error can be reset via CAN message. The control firmware will then be switched from state ERROR to state OFF:

<i>Identifier</i>	<i>DLC</i>	<i>Data</i>							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 _h	8	1F _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

Byte 1: Mode of operation Normal Mode (10_h) + Controlword QUIT_ERROR (0F_h)

Byte 3-4: End stops switched off

Byte 7: |Torque| ≤ 20% of the nominal torque

Byte 8: |Torque| ≤ 20% of the maximum torque

The SENSO-Stick responds with an acknowledging message with identifier 10_h:

<i>Identifier</i>	<i>DLC</i>	<i>Data</i>							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 _h	8	10 _h	XX _h	00 _h	00 _h	00 _h	00 _h	Enc-Idx	Enc-Idx

Byte 1-2: Mode of operation Normal Mode (10_h) + Controlword OFF (00_h)

Byte 3-4: Error

Byte 5-6: End stops switched off

Byte 7-8: Position of Encoder-Index

For a more detailed description of status and error messages please go to Chapter 6.2.3.

- 2) Switch the control firmware in the referencing mode to READY. Furthermore the end stops are deactivated and the torque limitation prepared at 100%:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	42 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	64 $_h$	64 $_h$

Byte 1: Referencing Mode (40 $_h$) + Controlword SWITCH_READY (02 $_h$)

Byte 3-4: End stops switched off

Byte 7: |Torque| $\leq$ 100% of nominal torque

Byte 8: |Torque| $\leq$ 100% of maximum torque

The SENSO-Stick responds with an acknowledging message with identifier 10 $\diamond_h$:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	42 $_h$	XX $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	Enc-Idx	Enc-Idx

Byte 1-2: Status

Byte 3-4: Error

Byte 5-6: End stops switched off

Byte 7-8: Position of Encoder-Index

A more detailed description of the referencing mode can be found in Chapter 6.5.

- 3) Start of Referencing: The joystick handle move autonomously when the control firmware is switched from READY to ON:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	44 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	64 $_h$	64 $_h$

Byte 1: Referencing Mode (40 $_h$) + Controlword SWITCH_ON (04 $_h$)

Byte 3-4: End stops switched off

Byte 7: |Drehmoment| $\leq$ 100% of nominal torque

Byte 8: |Drehmoment| $\leq$ 100% of maximum torque

The SENSO-Stick responds with an acknowledging message with identifier 10 $\diamond_h$:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	44 $_h$	XX $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	Enc-Idx	Enc-Idx

Byte 1-2: Status

Byte 3-4: Error

Byte 5-6: End stops switched off

Byte 7-8: Position of Encoder-Index

The handle of the SENSO-Sticks now moves slowly towards an end stop, remains there briefly and then moves toward the opposite end stop. After another pause in movement of the handle moves to the center position and remains there.

Referencing is now complete.

5.5 Step 3: Starting the SENSO-Stick

The torque is to be set to 20% for the first tests with an activated operation. In the following examples a torque limitation is already included (detailed in Chapter 6.2.2). With these settings the function of the SENSO-Stick can be safely tested. For normal operation values of 100% or higher are intended (see Chapter 6.2.2.6).

- 1) Switch the control firmware into Normal Mode:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	1F $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	14 $_h$	14 $_h$

Byte 1: Mode of operation Normal Mode (10 $_h$) + Controlword QUIT_ERROR (0F $_h$)

Byte 3-4: End stops switched off

Byte 7: |Torque| $\leq$ 20% of nominal torque

Byte 8: |Torque| $\leq$ 20% of maximum torque

The SENSO-Stick responds with an acknowledging message with identifier 10 $\diamond_h$:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	10 $_h$	XX $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	Enc-Idx	Enc-Idx

Byte 1-2: Status

Byte 3-4: Error

Byte 5-6: End stops switched off

Byte 7-8: Position of Encoder-Index

For a more detailed description of referencing mode please go to Chapter 6.5.

- 2) Switch the control firmware from state OFF to state READY, limit the torque to 20%:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	12 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	14 $_h$	14 $_h$

Byte 1: Mode of operation Normal Mode (10 $_h$) + Controlword SWITCH_READY (02 $_h$)

Byte 3-4: End stops switched off

Byte 7: |Torque| $\leq$ 20% of nominal torque

Byte 8: |Torque| $\leq$ 20% of maximum torque

The SENSO-Stick responds with an acknowledging message with identifier 10 $\diamond_h$:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	12 $_h$	XX $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	Enc-Idx	Enc-Idx

Byte 1-2: Status

Byte 3-4: Error

Byte 5-6: End stops switched off

Byte 7-8: Position of Encoder-Index

For a more detailed description of the status and error messages please go to Chapter 6.2.3.

3) Switch the control firmware from state READY to state ON:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	14 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	14 $_h$	14 $_h$

Byte 1: Mode of operation Normal Mode (10 $_h$) + Controlword SWITCH_ON (04 $_h$)

Byte 3-4: End stops switched off

Byte 7: |Torque| $\leq$ 20% of nominal torque

Byte 8: |Torque| $\leq$ 20% of maximum torque

The SENSO-Stick responds with the an acknowledging message with identifier 10 $\diamond_h$:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	14 $_h$	XX $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	Enc-Idx	Enc-Idx

Byte 1-2: Status

Byte 3-4: Error

Byte 5-6: End stops switched off

Byte 7-8: Position of Encoder-Index

For a more detailed description of status and error messages please go to Chapter 6.2.3.
The joystick handle should be possible to move freely around the zero position in the range $\pm 20^\circ$.

5.6 Step 4: Example of Parametrization of the SENSO-Stick

The SENSO-Stick is now operating in Normal Mode. An operation specific CAN message is used to parametrize the steering. For a basic simulation of the spring damping systems the following setting are used (Chapter 6.3):

Friction: 20 mNm = 0014 $_h$

Damping: 10 mNm/rpm = 000A $_h$

Spring stiffness: 10 mNm/1 $^\circ$ = 000A $_h$

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
21 $\diamond_h$	8	00 $_h$	00 $_h$	14 $_h$	00 $_h$	0A $_h$	00 $_h$	0A $_h$	00 $_h$

Byte 1-2: Desired torque

Byte 3-4: Friction

Byte 5: Damping

Byte 6: Alive-Counter

Byte 7-8: Spring stiffness

The self-aligning torque on the SENSO-Stick should rise with increasing deflection. By quicker movements the damping should be easily noticeable.

6 Description of the Control Firmware

This chapter describes the firmware modules of the SENSO-Stick control firmware. The host controller (host) is connected to the SENSO-Stick controller by a 1Mbit/s CAN connection. Except Referencing each of the modules has two CAN Identifiers: One identifier to receive messages from the host and one to respond to the host.

6.1 Overview of the CAN-Messages of the SENSO-Stick

The following table shows the reserved CAN identifiers (ID) for the SENSO-Stick.

CAN messages that are sent to the SENSO-Stick have a gray background. White lines/fields contain responses from the SENSO-Stick.

<i>ID x-Axis</i>	<i>ID y-Axis</i>	<i>ID z-Axis</i>	<i>Direction</i>	<i>Description</i>	<i>Page</i>
011 _h	012 _h	013 _h	Host → Stick ¹	Firmware-Update	
101 _h	102 _h	103 _h	Stick → Host	Control module status	21
201 _h	202 _h	203 _h	Host → Stick	Control module	21
111 _h	112 _h	113 _h	Stick → Host	Normal mode actual readings	28
211 _h	212 _h	213 _h	Host → Stick	Normal Mode desired values	28
121 _h	122 _h	123 _h	Stick → Host	Basic mode actual readings	29
221 _h	222 _h	223 _h	Host → Stick	Basic Mode desired values	29
131 _h	132 _h	133 _h	Stick → Host	Referencing, actual readings	30
231 _h	232 _h	233 _h	Host → Stick	Referencing, desired values	30
141 _h	142 _h	143 _h	Stick → Host	SENSODRIVE Demonstration mode actual readings	
241 _h	242 _h	243 _h	Host → Stick	SENSODRIVE Demonstration mode desired values	
151 _h	152 _h	153 _h	Stick → Host	Digital and analog inputs	37
251 _h	252 _h	253 _h	Host → Stick	Set digital outputs, request inputs	37
161 _h	162 _h	163 _h	Stick → Host	Confirmation of Absolute Position	34
261 _h	262 _h	263 _h	Host → Stick	Set the Absolute Position	34
171 _h	172 _h	173 _h	Stick → Host	Read-only values	44
271 _h	272 _h	273 _h	Host → Stick	Request read-only values	44
181 _h	182 _h	183 _h	Stick → Host	Version information	46
281 _h	282 _h	283 _h	Host → Stick	Request version information	46
1A1 _h	1A2 _h	1A3 _h	Stick → Host	Parameters for SENSO-Stick effects	40
2A1 _h	2A2 _h	2A3 _h	Host → Stick	Set parameters for SENSO-Stick effects	40
1B1 _h	1B2 _h	1B3 _h	Stick → Host	System parameters	43
2B1 _h	2B2 _h	2B3 _h	Host → Stick	Request system parameters	43
1F1 _h	1F2 _h	1F3 _h	Stick → Host	Manufacturer parameters	
2F1 _h	2F2 _h	2F3 _h	Host → Stick	Manufacturer parameters	

¹ SENSO-Stick

6.2 Control Module

The control module is the central control unit of the SENSO-Stick. It computes common tasks for all other modules and it activates and deactivates the different modes of operation.

6.2.1 State Machine of the Control Firmware

The control module of the SENSO-Stick implements a state machine with the four states OFF, READY, ON, and ERROR.

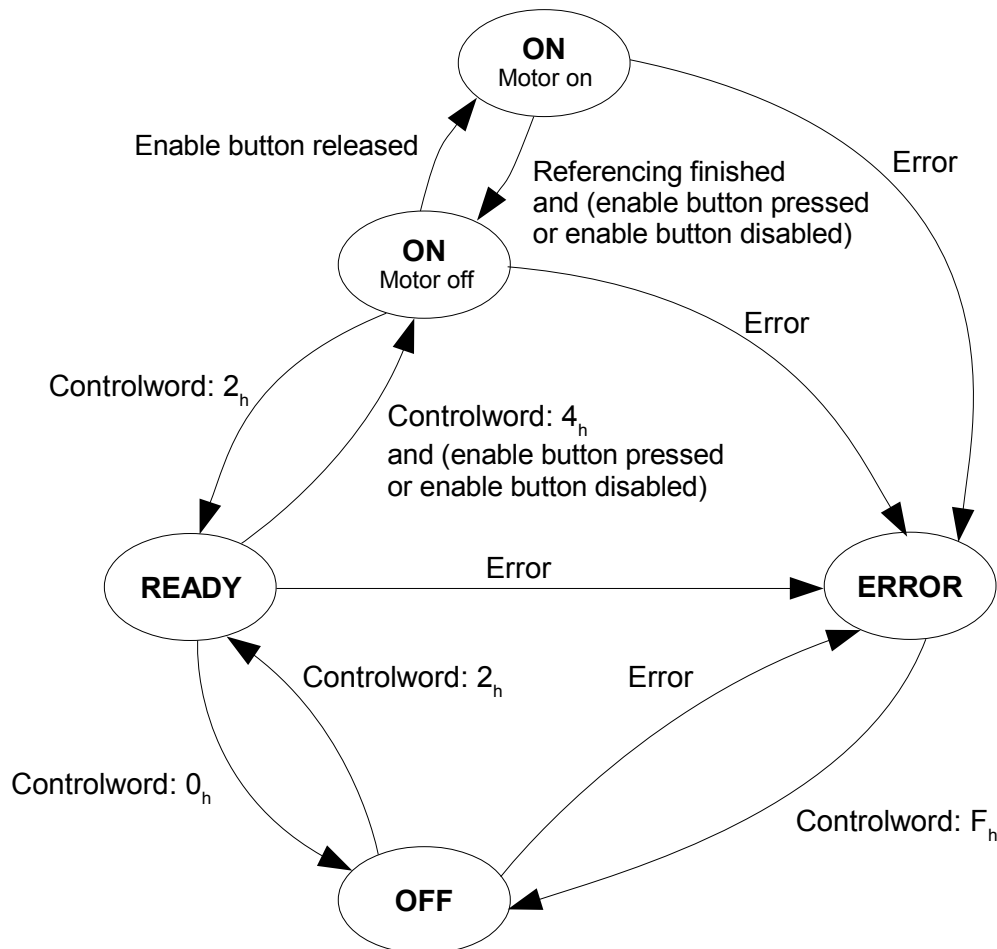


Fig. 6-1: State machine of the control firmware

Restarting the controller resets the control firmware to state OFF. The SENSO-Stick will transit to state READY after receiving the controlword 2_h from the host. The control firmware goes from state READY to state ON when the controlword 4_h is sent and at the same time either when the enable button is pressed or deactivated.

The motor is only in state ON by pressing or by deactivated enable button and a successfully completed referencing.

If an error occurs, the control firmware will switch to state ERROR. The reason for the error has to be removed first and the error has to be acknowledged by the host with controlword F_h , before the control software transits to state OFF. The motor can then be re-activated as stated above.

6.2.2 Functions of the Control Module

Control Module				
Identifier: 20 _h				
DLC: 8 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Controlword		0 _h , 2 _h , 4 _h , F _h	1	0 ... 3
Mode of Operation		1 _h , 2 _h , 4 _h , 8 _h	1	4 ... 7
Auxiliary Functions		0 _h - FF _h	2	0 ... 7
End Stop Positions	[1°]	1° - 90°	3, 4	0 ... 15
Position Offset	[1°]	±359°	5, 6	0 ... 15
Torque Limitation	[1%]	0% - 255%	7	0 ... 7
Peak Torque Limitation	[1%]	0% - 100%	8	0 ... 7

The receiving protocol contains all parameter settings for the SENSO-Stick. These parameters are described in more detail in the following chapters. There are control module functions without parameters. These functions are described as well.

6.2.2.1 Controlword

The controlword triggers a change in state of the state machine. It is placed in the four least significant bits of the first Byte.

Controlword	
0 _h	SWITCH_OFF
2 _h	SWITCH_READY
4 _h	SWITCH_ON → activate motor
F _h	QUIT_ERROR

6.2.2.2 Mode of Operation

Mode of Operation selects one of four different SENSO-Stick modes. It is placed in the four most significant bits of the first Byte. Changing the Mode of Operation in state ON will cause a transition to state OFF.

Mode of Operation	
1 _h	Normal Mode
2 _h	Basic Mode
4 _h	Referencing Mode
8 _h	Demonstration Mode

6.2.2.3 Auxiliary Functions

Several auxiliary functions can be chosen bit by bit in Byte 2.

<i>Auxiliary Functions</i>		
00 _h	0000 0000 _b	default
01 _h	0000 0001 _b	CAN Watchdog off
08 _h	0000 1000 _b	Enable button deactivated
10 _h	0001 0000 _h	Alive-Counter deactivated

CAN Watchdog

Each of the SENSO-Stick controller expects a message from the host every 50ms (5ms in Basic Mode). If the message does not arrive on time, the CAN Watchdog switches off the motor. The controller software responds with the error message "CAN Watchdog Timeout" 0080_h and enters the state ERROR.

The CAN Watchdog is activated in state ON (Motor activated). The CAN Watchdog can be switched off by the user under certain circumstances, e.g. while the SENSO-Stick is connected to the host. Only qualified personnel is allowed to operate the SENSO-Stick while the CAN Watchdog is switched off.



During operation the CAN Watchdog must be switched on at all times!

Enable Button

The motors of the SENSO-Sticks are only activated in the state ON in the control firmware when the enable button is pressed on the joystick handle. Once the enable button is released, the motors are switched off. As shown in Fig. 6-1, the state machine control firmware remains in state ON.

In special cases (e.g. as optimization of parameters), it may be useful to disable the enable button. Only qualified personnel is allowed to operate the SENSO-Stick while the enable button is disabled.

Alive Counter

The alive counter is used for the mutual monitoring of the SENSO-Stick and host system. For this purpose, with each normal mode message a counter variable is sent (see Chapter 6.3).

Counting from 0 to 15, then again starting at 0. The order must be monitored both on the SENSO-Stick and in the host system. With each transmitted normal mode message the host system increases the alive counter value by 1. The SENSO-Stick returns the received Alive-Counter value.

The Alive-Counter is only available in the normal operation mode. The alive counter starts automatically with the value 0. By an Error reset the alive counter is also reset to 0.

By default, the alive counter is activated. With the auxiliary function 10_n the alive counter can be switched off.

6.2.2.4 End Stops

End stops are available in mode of operation Normal Mode. The end stop positions can be modified. In case the chosen end stop positions are less than 1° the end stops will be deactivated. The end stop position is limited to 25°.

6.2.2.5 Position Offset

The SENSO-Stick does not have access to an absolute position. After powering-up the position output is always zero, independent of the position of the joystick.

Nevertheless, in order for a defined zero position the operation mode referencing has to be used. The parameter "position offset" is used only in the operation mode referencing. A detailed description of this mode can be found in Chapter 6.5.

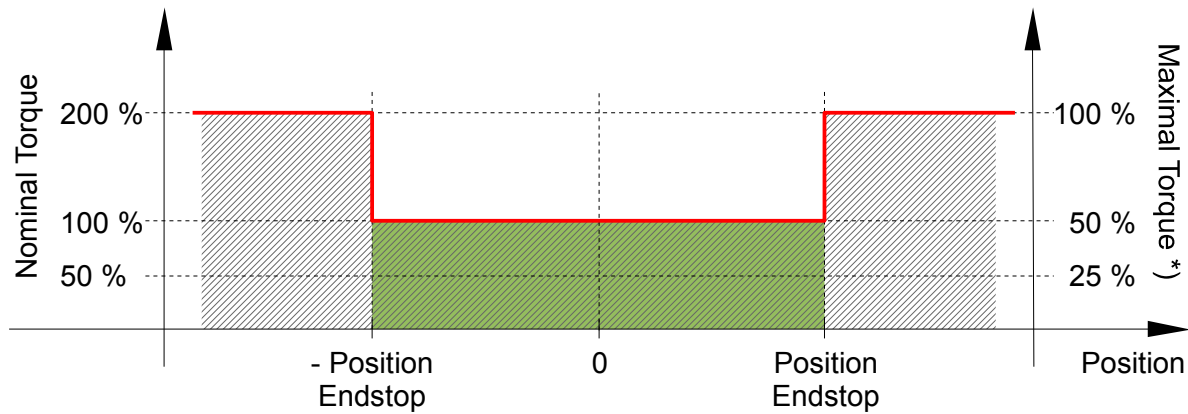
6.2.2.6 Torque Limitation

The actual torque can be limited e.g. for the initial set-up or different safety reasons. The parameters "Torque Limitation" and "Peak Torque Limitation" are provided for that.

The parameter "Peak Torque Limitation" defines the maximum torque at the end stops ($|position| > position \text{ of end stops}$). This parameter is given as a percentage of the maximum torque of the motor.

The parameter "Torque Limitation" gives the maximum value of the internally calculated desired torque (= combination of friction, spring stiffness, desired torque, ...) as a percentage of the nominal torque of the motor. This limit is valid for positions within the end stops ($|position| < position \text{ of end stops}$, see Fig. 6-2, green area). In order to be able to output the maximum torque of the motor between the end stops, the parameter "Torque Limitation" can be set to >100%.

The examples and Fig. 6-2 are explaining the case in which the motor maximum torque is exactly twice as much as the motor nominal torque. If the difference is higher, the "Torque Limitation" may be increased up to 254%. A value of 255% means in any case, that "Torque Limitation" equals "Peak Torque Limit".



*) in this example: Maximal Torque = 2x Nominal Torque

Fig. 6-2: Torque limitation

The limited peak torque is always greater or equal to the limited torque and will always avoid a lower torque at the end stops than in between. For example if the “Torque Limitation” is set to 50%, then the “Peak Torque Limitation” must be set to $\geq 25\%$ (indicated by a dashed line in Fig. 6-2). A given lower value of Peak Torque Limit is adjusted in the software accordingly.

6.2.2.7 Activating the Motor when $|\text{Position}| > \text{Position of End Stops}$

In case the handle is in a position beyond the end stops ($|\text{position}| > \text{position of end stops}$), the SENSO-Stick motor cannot be activated (transition from state READY to state ON). Activating the motor beyond the end stops would cause the torque to jump from 0Nm to peak torque, which would be a potential risk. In that case the controller does not activate the motor, enters the state ERROR and posts the error message 0200_h “Switched on while $|\text{Position}| > \text{End Stop Position}$ ”.

6.2.2.8 Overload Protection for the Motor

The motor can provide maximum torque without overheating for a short time only. Hence the controller software limits the actual torque to the nominal torque after a few seconds. The timeframe until the reduction is activated depends on the overload. A smaller overload prolongs the allowable overload time.

6.2.2.9 Test of CPU Watchdog

The controller CPU has a built-in CPU watchdog for monitoring. Every time the controller is powered-up this watchdog is tested. After a successful check the status message “Test CPU-Watchdog successful” is posted.

6.2.3 Status SENSO-Stick

Whenever the control module receives a message from the host, an interrupt service routine saves the message for further processing. The routine also replies to the host with status, error, end stop positions and the encoder index position. This reply does not include the results from the current message. This can only be seen in the succeeding message.

Control Module Status				
Identifier: 10 $\diamond$ _h				
DLC: 8 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Status	-	0 - FFFF _h	1, 2	0 ... 15
Error	-	0 - FFFF _h	3, 4	0 ... 15
End Stop Position	[0.1°]	0 - 250	5, 6	0 ... 15
Encoder Index Position	[1°]	±359	7, 8	0 ... 15

The status is hexadecimal coded. The lower four bits contain the information about the current state of the state machine. The next four bits inform about the actual mode of operation. The upper 8 bits provide general status information.

Description Status Message	
0000 _h	State OFF
0002 _h	State READY
0004 _h	State ON → Motor activated
0008 _h	State ERROR
0010 _h	Mode of Operation: Normal Mode
0020 _h	Mode of Operation: Basic Mode
0040 _h	Mode of Operation: Referencing Mode
0080 _h	Alive-Counter Activated
0100 _h	CAN Watchdog Off
0200 _h	Overload Protection Active
0400 _h	End Stops Active
0800 _h	Encoderindex
1000 _h	Enable Button Off
2000 _h	Referencing Completed
4000 _h	Test CPU Watchdog successful

Usually the status message is a combination of more messages. The individual messages are added.

Example: Status message 0C14_h is a combination of 0800_h, 0400_h, 0010_h und 0004_h.
 0004_h: State ON → Motor Active
 0010_h: Mode of operation Normal Mode
 0400_h: End Stops Active
 0800_h: Encoder index

The error message is hexadecimal coded as well and more messages can be combined into one.

Description Error Message	
0001 _h	Overcurrent in Output Stage
0002 _h	Supply Voltage too High
0004 _h	Supply Voltage too Low
0008 _h	Overtemperature in Output Stage
0010 _h	Hall Sensor Fault ¹
0020 _h	CAN Bus Error
0040 _h	Hall Sensors swapped / Encoder Fault ¹
0080 _h	CAN-Watchdog Timeout
0100 _h	Illegal Position (Position > 90°)
0200 _h	Switched on while Position > End Stop Position
0400 _h	Referencing Timeout
0800 _h	Overtemperature in Brake Chopper
1000 _h	Error Alive-Counter (Difference > 1)
2000 _h	Collective Error (see Chapter 7.4)
4000 _h	Wrong Number of Firmware Update Messages
8000 _h	Firmware Update Checksum Error

¹ Error in hardware. These errors cannot be quit with the controlword via CAN! Otherwise, the position actual value may change unexpectedly!

6.3 Normal Mode

The control module takes care of general tasks while the modules for the individual operating modes perform specialized processes.

The SENSO-Stick is in normal mode. It then acts as a force feedback joystick. The controller firmware cycle is 1ms. Four parameters can be adjusted to create a realistic feeling: desired torque, friction, damping and spring stiffness. The resulting motor torque is calculated as:

Motor torque = desired torque + friction + damping · velocity + spring stiffness · position.

The parameters are transmitted in a message with identifier 21 $\diamond_h$ to the SENSO-Stick. The SENSO-Stick replies with a message with identifier 11 $\diamond_h$ with the current values of position, velocity and torque. The position is transmitted as number of increments. The total number of increments per revolution can be read via CAN (refer to Chapter 7.3).

Consequently this leaves the handle position to be converted in degrees [°]:

$$\text{Handle Positon}[^{\circ}] = \frac{\text{Handle Positon}[\text{inc}]}{\text{Position increments/revolution}[\text{inc/rev}]} \cdot 360[^{\circ}/\text{rev}] \quad (1)$$

Desired Values for Normal Mode				
Identifier: 21 $\diamond_h$				
DLC: 8 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Desired Torque	[mNm]	-32768...32767 ¹	1, 2	0 ... 15
Friction	[mNm]	0 - 500	3, 4	0 ... 15
Damping	[mNm/rpm]	0 - 50	5	0 ... 7
Alive-Counter	[-]	0 - 15	6	0 ... 7
Spring Stiffness	[mNm/1°]	0 - 2500	7, 8	0 ... 15

The resulting motor torque is calculated in the 1ms cycle and sent to the current controller.

Actual Readings for Normal Mode				
Identifier: 11 $\diamond_h$				
DLC: 8 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Handle Position	[inc]	-2 ²³ ... 2 ²³ -1	1-3	0 ... 23
Alive-Counter	[-]	0 - 15	4	0 ... 7
Velocity	[rpm]	-32768 ... 32767	5, 6	0 ... 15
Actual Torque	[mNm]	-32768 ... 32767 ¹	7, 8	0 ... 15

The values of position and actual torque are updated every 50μs, the value of velocity every 1ms. To derive the velocity, the difference of the current position and the position of 1ms ago is calculated and filtered. Therefore the velocity value is the average velocity of the past 1ms.

¹ For torque limits please refer to chapter 6.2.2 and the technical data in the operating manual of the SENSO-Stick.

6.4 Basic Mode

The Basic Mode is designed for applications with a cycle time less than 1ms, but more than 300µs. To achieve the fast cycle times the work load on the SENSO-Stick controller and the traffic on the CAN bus have to be reduced, thus enabling the high cycle times, only the nominal torque can be specified. As the current value of the SENSO-Stick returns only the position.

The end stops are switched off. The parameter "Torque Limitation" (refer to chapter 6.2.2) may be set to 255% to use the maximum torque of the motor.

Betriebsart Basismodus Sollwerte				
Identifier: 22 _h				
DLC: 2 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Desired Torque	[mNm]	-32768...32767 ¹	1, 2	0 ... 15

The desired torque is passed to the current controller every 50µs.

Actual Readings for Basic Mode				
Identifier: 12 _h				
DLC: 4 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Position	[inc]	-2 ²³ ... 2 ²³ -1	1, 2, 3	0 ... 23

The position value is updated every 50µs.

¹ For torque limits please refer to chapter 6.2.2 and the technical data in the Operating Manual of the SENSO-Stick.

6.5 Referencing Mode

The SENSO-Stick does not come with an absolute position signal. This means that after powering-up, the position output is always zero regardless of the joystick handle position during powering-up.

Therefore after powering-up the SENSO-Stick has to be referenced first. This can be done by a referencing run to search for the end stops (Chapter 6.5.1). The SENSO-Stick can be given an absolute position via CAN (Chapter 6.5.2).

The zero position that was referenced by applying one of these two options i.e. the zero position of the SENSO-Stick can automatically be approached by homing at any time (Chapter 6.5.3).

Due to the design of the mechanism the transmission changes with movement of the joystick handle. To compensate for this variable transmission it is essential that a referencing of the SENSO-Stick is carried out. This also ensures that the effects and torque of the handle correspond to the commands.

6.5.1 Referencing to Mechanical End Stops

6.5.1.1 Automatic Referencing

By automatic referencing the end stops on both sides of the chosen axis are automatically hit. After referencing the zero position in the middle between the end stops is set. Referencing is done with the Control Module (CAN message 20_h). The controlword 40_h in Byte 1 sets the SENSO-Stick mode to referencing. Byte 1 of the message is set according to the state machine first at READY then at ON (Controlword 42_h and 44_h). The referencing begins. After completion the SENSO-Stick moves automatically in the determined position and remains there.

The gradual approach should be exemplified here:

- 1) SENSO-Stick is shifted into operation mode "Referencing", state READY by a message of the control module, taking into account the transitions allowable according to the finite state machine, the Position offset is set to 0°:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 _h	8	42 _h	00 _h	00 _h	00 _h	00 _h	00 _h	64 _h	64 _h

Byte 1: Mode of Operating: Referencing (40_h) + Controlword "SWITCH_READY" (02_h)

Byte 2-4: Irrelevant for referencing

Byte 5-6: Position offset correction

Byte 7: Nominal torque limitation at 100%

Byte 8: Peak torque limitation at 100%

The SENSO-Stick responds with the following message:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 _h	8	XX _h	XX _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

Byte 1-2: Status with preset mode of operation and state

Byte 3-4: Error

Byte 5-6: Position of end stops

Byte 7-8: Position of encoder index

- 2) Switched on in the state ON the joystick begins to move slowly in a direction. As soon as an end stop is recognized the SENSO-Stick changes its direction. After approaching the second end stop the joystick moves automatically to the middle position and remains there.

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	44 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	64 $_h$	64 $_h$

Byte 1: Mode of operation: referencing (40 $_h$) + Controlword "SWITCH_ON" (04 $_h$)

Byte 2-4: Irrelevant for Referencing

Byte 5-6: Position offset correction (e.g. 0005 $_h$, corresponds with $\phi_{idx} = 5^\circ$)

Byte 7: Nominal torque limitation at 100%

Byte 8: Peak torque limitation at 100%

- 3) The answers on further CAN messages 20 $\diamond_h$ as under in Point 1) show in the status that it's the End of Referencing:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	44 $_h$	68 $_h$	00 $_h$	00 $_h$	B4 $_h$	00 $_h$	00 $_h$	00 $_h$

Byte 1-2: Status = 6844 $_h$

Byte 3-4: Error

Byte 5-6: Position of End Stops

Byte 7-8: Position of Encoder-Index (irrelevant for referencing on Mechanical Endstops)

The maximum torque the controller can command in operation mode Referencing is limited to 400mNm. Velocity is equally limited. If the joystick handle doesn't go to the end position within 30 seconds it switches to state ERROR, issues a failure message (0400 $_h$, "Referencing Timeout") and shuts down the motor.

6.5.1.2 Manual Referencing

In addition to automatic referencing it is also possible to reference the SENSO-Stick manually to the mechanical end stop. Reset/turn on the SESO-Stick. The SENSO-Stick is in state OFF. In Control Module (CAN message 20 $\diamond_h$) the mode of operation: Referencing 40 $_h$ + controlword READY 02 $_h$ is set. In addition the host must send the CAN message referencing (23 $\diamond_h$) and the referencing controlword 20 $_h$. At this point the SENSO-Stick manually moves to the two mechanical end stops. With the CAN message (23 $\diamond_h$) and the controlword 21 $_h$ the end stop is set. If the state now changes to ON, the SENSO-Stick moves autonomously to the middle position.

The gradual approach should be exemplified here:

- 1) SENSO-Stick is shifted into operation mode "Referencing", state READY by a message of the control module, taking into account the transitions allowable according to the finite state machine, the Position offset is set (only if not referenced to absolute position) as following:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 $\diamond_h$	8	42 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	64 $_h$	64 $_h$

- Byte 1: Mode of Operation Referencing (40 $_h$) + Controlword "SWITCH_READY" (02 $_h$)
- Byte 2-4: irrelevant for referencing
- Byte 5-6: Position offset correction
- Byte 7: Nominal torque limitation at 100%
- Byte 8: Peak torque limitation at 100%

The SENSO-Stick responds with the following message:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond_h$	8	XX $_h$	XX $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$

- Byte 1-2: Status with the preset mode of operation and state
- Byte 3-4: Error
- Byte 5-6: Position of end stops
- Byte 7-8: Position of Encoder-Index

- 2) With help of the CAN message 23 $\diamond_h$ the type of referencing is adapted. Through setting the first byte to 20 $_h$ the type is set to manual:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
23 $\diamond_h$	8	20 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$	00 $_h$

- Byte 1-2: Mode of Operation: Manual (20 $_h$) + Controlword (00 $_h$)
- Byte 3-8: irrelevant

The SENSO-Stick responds with the following message:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
13 $\diamond_h$	8	21 $_h$	00 $_h$	XX $_h$	XX $_h$	XX $_h$	XX $_h$	00 $_h$	00 $_h$

- Byte 1-2: Status with the preset mode of operation and state
- Byte 3-6: Position of Motor
- Byte 5-6: Motor torque

- 3) If all end stops were approached manually with the joystick handle, the mean value can be set with the controlword 01_h:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
23 _h	8	21 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

Byte 1-2: Referencing type: manual (20_h) + controlword (01_h) to set the position

Byte 3-8: irrelevant

The SENSO-Stick responds with the following message:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
13 _h	8	21 _h	00 _h	XX _h	XX _h	XX _h	XX _h	00 _h	00 _h

Byte 1-2: Status with the preset mode of operation and state

Byte 3-6: Position of motor

Byte 5-6: Motor torque

- 4) Switch in the state ON: The joystick moves automatically to the middle position and remains there.

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 _h	8	44 _h	00 _h	00 _h	00 _h	05 _h	00 _h	64 _h	64 _h

Byte 1: Mode of operation Referencing (40_h) + control word "SWITCH_ON" (04_h)

Byte 2-4: Irrelevant for Referencing

Byte 5-6: Position offset correction (e.g. 0005_h, corresponds with $\phi_{idx} = 5^\circ$)

Byte 7: Nominal torque limitation at 100%

Byte 8: Peak torque limitation at 100%

- 5) The answers from further CAN messages 23_h as below in point 1) shows in Status the End of Referencing:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 _h	8	44 _h	68 _h	00 _h	00 _h	B4 _h	00 _h	00 _h	00 _h

Byte 1-2: Status = 6844_h

Byte 3-4: Error

Byte 5-6: Position of end stops

Byte 7-8: Position of Encoder-Index (irrelevant for referencing on Mechanical Endstops)

6.5.2 Referencing to Absolute Position

If there is an absolute position of the system, then the SENSO-Stick position can be specified. The absolute position information can be read, for example, from an external digital sensor or a potentiometer in the master controller and ultimately sent via CAN to the SENSO-Stick. This then takes the received position as its own current position.

A potentiometer can be read e.g., through the 10bit analog input of the SENSO-Stick per CAN in the motor controller (only for modular SENSO-Stick).

Setting of Absolute Position				
Identifier: 26 $\diamond_h$				
DLC: 2 Byte				
Direction: Host $\rightarrow$ SENSO-Stick				
	Unit	Range	Byte	Bits
Absolute Position	[0.1°]	+/- 14400	1, 2	0 ... 15

The CAN message for setting the absolute position is only acknowledged and responded to if SENSO-Stick is not in state ON.

Absolute Position Confirmation				
Identifier: 16 $\diamond_h$				
DLC: 2 Byte				
Direction: SENSO-Stick $\rightarrow$ Host				
	Unit	Range	Byte	Bits
Absolute Position	[0.1°]	+/- 14400	1, 2	0 ... 15

The procedure will be explained by way of example here:

Identifier	DLC	Data	
		<i>Byte 1</i>	<i>Byte 2</i>
261 $_h$	2	B6 $_h$	03 $_h$

Byte 1-2: Absolute Position (e.g. 03B6 $_h$, corresponds with 95.0°)

The SENSO-Stick responds with the following message:

Identifier	DLC	Data	
		<i>Byte 1</i>	<i>Byte 2</i>
161 $_h$	2	B6 $_h$	03 $_h$

Hence the actual value of the SENSO-Stick is now 95.0°.

Referencing to an index is not possible anymore after setting the absolute position, apart from switching the SENSO-Stick on and off again (Reset). A zero position is still possible via the Control Module (CAN message 20 $\diamond_h$) with the mode of operation: Referencing (Chapter 6.5.3).

6.5.3 Homing

The SENSO-Stick is in an arbitrary position after switching it on. If it isn't referenced on an absolute position, (Chapter 6.5.2) the referencing of the mechanical end stops will be performed once it has been switched into the mode of operation: referencing automatic. (Chapter 6.5.1). The zero position for homing can be corrected through the position-offset in the Control Module (CAN message 20_h).

If from another operating mode the zero drive is started again, or it has been referenced to an absolute position, the desired zero position is approached directly, without referencing to the mechanical end stops again.

The gradual approach is exemplified here:

- 1) SENSO-Stick is shifted into operation mode "Referencing", state READY by a message of the control module, taking into account the transitions allowable according to the finite state machine, the Position offset is set (only if not referenced to absolute position) as following:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 _h	8	42 _h	00 _h	00 _h	00 _h	05 _h	00 _h	64 _h	64 _h

Byte 1: Mode of operation referencing (40_h) + control word "SWITCH_READY" (02_h)

Byte 2-4: Irrelevant for Referencing

Byte 5-6: Position offset correction (e.g. 0005_h, corresponds with $\varphi_{idx} = 5^\circ$)

Byte 7: Nominal torque limitation at 100%

Byte 8: Peak torque limitation at 100%

The SENSO-Stick responds with following message:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 _h	8	XX _h	XX _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

Byte 1-2: Status with preset mode of operation and state

Byte 3-4: Error

Byte 5-6: Position of end stops

Byte 7-8: Position of encoder index

- 2) Switched to state ON: The joystick handle slowly begins to move.

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
20 _h	8	44 _h	00 _h	00 _h	00 _h	05 _h	00 _h	64 _h	64 _h

Byte 1: Mode of operation referencing (40_h) + control word "SWITCH_ON" (04_h)

Byte 2-4: Irrelevant for Referencing

Byte 5-6: Position offset correction (e.g. 0005_h, corresponds with $\varphi_{idx} = 5^\circ$)

Byte 7: Nominal torque limitation at 100%

Byte 8: Peak torque limitation at 100%

- 3) As soon as the zero position is achieved the joystick handle stops. The answers on further CAN messages 20 $\diamond$ _h (as below in point 2) show in the Status the end of Referencing:

Identifier	DLC	Data							
		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
10 $\diamond$ _h	8	44 _h	68 _h	00 _h	00 _h	B4 _h	00 _h	05 _h	00 _h

Byte 1-2: Status = 6844_h

Byte 3-4: Error

Byte 5-6: Position of end stops

Byte 7-8: Position of encoder index (e.g. 0005_h, corresponds with $\varphi_{idx} = 5^\circ$)

The maximum torque that can be commanded in operation mode Referencing is limited to 400 mNm. Velocity is limited as well. If the joystick handle doesn't go to the end position within 30 seconds it switches to state ERROR, issues a failure message (0400_h, "Referencing Timeout") and shuts down the motor.

6.5.4 Reset and repeat Referencing

Referencing can only be carried out once. A reset or repeat referencing is not envisaged for the SENSO-Stick. If you want to carry out a new referencing, then the SENSO-Stick has to be briefly disconnected from the power supply.

6.6 Reading of Buttons und Switches on the Joystick Handle

The buttons and switches can be read via the digital inputs and outputs of the three built-in SENSO-Stick motor controllers.

A further documentation of the digital inputs and outputs is given in Chapter 7.1.

To retrieve the switch status of the button on the joystick handle the following CAN message is sent:

Retrieving Switch Status
Identifier: 25 $\diamond_h$
DLC: 0 Byte
Direction: Host → SENSO-Stick

The SENSO-Stick responds with the actual values of the button.

Return Switch Status				
Identifier: 15 $\diamond_{\text{h}}$				
DLC: 6 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Digital Inputs	-	0 – 003F _h	1, 2	0 ... 15
Analog Input 1 ¹		0 – 03FF _h	3, 4	0 ... 15
Analog Input 2 ¹		0 – 03FF _h	5, 6	0 ... 15

The following table shows the assignment of buttons and switches to the digital inputs:

Assignment of Message – pressing button			
Value	Controller		
	x-Axis , ID: 151 $_h$	y-Axis , ID: 152 $_h$	z-Axis , ID: 153 $_h$
00 $_h$	None of the following buttons were pressed	None of the following buttons were pressed	None of the following buttons were pressed
01 $_h$	Rear button (enable button)	Rear button (enable button)	Rear button (enable button)
02 $_h$	Ministick, left	Flip switch, left	- not assigned -
04 $_h$	Ministick, right	Flip switch, right	- not assigned -
08 $_h$	Ministick, up	Front button (Push)	- not assigned -
10 $_h$	Ministick, down	- not available -	- not available -
20 $_h$	Ministick, pressed	- not available -	- not available -

The values are combined bit by bit, for example:

Value 0B $_h$ enable button (01 $_h$) + flip switch left (02 $_h$) + front button (Push) (08 $_h$)

¹ not by the controller of the x axis

7 Additional Functions of the SENSO-Stick Firmware

7.1 Analog and Digital I/O

This feature is not available for the integrated version of the SENSO-Stick because the inputs and outputs are not accessible.

The function of the digital inputs and outputs can also be used for reading the buttons and switches on the joystick handle (see Chapter 6.6). Some inputs are therefore already assigned and can no longer be used.

The following CAN message can be used to set the digital outputs and to read the digital and analog inputs. The digital outputs can be set bit by bit.

Digital Output Values				
Identifier: 25 $\diamond$ _h				
DLC: 2 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Digital Outputs	-	00 _h - 07 _h	1, 2	0 ... 15

The digital outputs are coded as follows:

- Value 00_h all outputs are reset
- Value 01_h Output 1 is set
- Value 02_h Output 2 is set
- Value 04_h Output 3 is set

The values are combined bit by bit, for example:

- Value 06_h Output 1 is not set, outputs 2 and 3 are set.

The SENSO-Stick responds with the current values of the analog and digital inputs. Available digital inputs depend on the axis (see above).

Digital and Analog Input Values				
Identifier: 15 _h				
DLC: 6 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Digital Inputs	-	0 – 003F _h	1, 2	0 ... 15
Analog input 1 ¹		0 – 03FF _h	3, 4	0 ... 15
Analog input 2 ¹		0 – 03FF _h	5, 6	0 ... 15

The digital inputs are coded as follows:

Value 00 _h	all inputs LOW
Value 01 _h	Input 1 HIGH
Value 02 _h	Input 2 HIGH
Value 04 _h	Input 3 HIGH
Value 08 _h	Input 4 HIGH
Value 10 _h	Input 5 HIGH (only by the controller of the x-axis)
Value 20 _h	Input 6 HIGH (only by the controller of the x-axis)

The values are combined bit by bit, for example:

Value 16_h Inputs 1, 4, and 6 are logic LOW, inputs 2, 3, and 5 are logic HIGH.

The voltage range of the analog inputs is 0...+4.8V, where 4.8V corresponds to 03FF_h.

¹ not by controller of the x-axis

7.2 Optional Parametrization of the SENSO-Stick Effects

To avoid vibrations, the default parameters may be changed in both Normal Mode and Referencing Mode by optional CAN messages. These values will not be saved, i.e. they have to be send to the SENSO-Stick again after powering up.



Changing the parameters may reduce the effect (e.g. softer end stops).

The parameters are merged to data sets using one collective CAN-Identifier. The first byte will distinguish different data sets.

Byte 1	Byte 2	Byte 3, 4	Byte 5, 6	Byte 7, 8
Data set	-	Variable 1	Variable 2	Variable 3
01 _h	0	End stops – Stiffness	End stops – Damping	0
		Unit: [mNm/1°]	Unit: [mNm/rpm]	
		Default: 150	Default: 10	
02 _h	0	Referencing – Damping	0	0
		Unit: [mNm/rpm]		
		Default: 30		
05 _h	0	Friction (Version 2) – Damping	Friction (Version 2) – Factor	0
		Unit: [mNm/rpm]	Unit: [%]	
		Default: 400	Default: 100	
06 _h	0	Damping – Velocity at which the half damping value is reached	Damping – Velocity filter constant k_{T1} for $T_1 = 65.535s / k_{T1}$	0
		Unit: [0.1 rpm]	Unit: 1	
		Default: 0	Default: 5000	
		Range: [0 ... 1000]	Range: [1 ... 65535]	

To avoid vibrations these values may be changed.

The CAN message for parametrization is as follows:

Setting optional parameters of SENSO-Stick effects				
Identifier: 2A $\diamond_h$				
DLC: 8 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Data set	-	1, 2, 5, 6	1	0 ... 7
reserved			2	0 ... 7
Variable 1		n. limited	3, 4	0 ... 15
Variable 2		n. limited	5, 6	0 ... 15
Variable 3		n. limited	7, 8	0 ... 15

The SENSO-Stick responds the actual values:

Optional parameters of SENSO-Stick effects				
Identifier: 1A $\diamond_h$				
DLC: 8 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Data set	-	1, 2, 5, 6	1	0 ... 7
reserved	-		2	0 ... 7
Variable 1		n. limited	3, 4	0 ... 15
Variable 2		n. limited	5, 6	0 ... 15
Variable 3		n. limited	7, 8	0 ... 15

The actual values may be read specifying the data set:

Request optional parameters of SENSO-Stick effects				
Identifier: 2A $\diamond_h$				
DLC: 1 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Data set	-	1, 2, 5, 6	1	0 ... 7

Example for setting the end stops (Values: Stiffness = 75, Damping = 5):

<i>Identifier</i>	<i>DLC</i>	<i>Data</i>							
		<i>Byte 1</i>	<i>Byte 2</i>	<i>Byte 3</i>	<i>Byte 4</i>	<i>Byte 5</i>	<i>Byte 6</i>	<i>Byte 7</i>	<i>Byte 8</i>
2A _h	8	01 _h	00 _h	4B _h	00 _h	05 _h	00 _h	00 _h	00 _h

Example to request the actual values for referencing (Data set 2):

<i>Identifier</i>	<i>DLC</i>	<i>Data</i>
		<i>Byte 1</i>
2A1 _h	1	02 _h

Example for an answer (Value: Damping = 30):

<i>Identifier</i>	<i>DLC</i>	<i>Data</i>							
		<i>Byte 1</i>	<i>Byte 2</i>	<i>Byte 3</i>	<i>Byte 4</i>	<i>Byte 5</i>	<i>Byte 6</i>	<i>Byte 7</i>	<i>Byte 8</i>
1A1 _h	8	1E _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h	00 _h

7.3 System Parameters

To read characteristic parameters from the system, the following CAN message can be used. For example, the number of position increments per revolution may be useful to calculate the position in degree [°].

<i>Request system parameters</i>				
Identifier: 2B $\diamond_h$				
DLC: 1 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Data set	-	1: Position incs. 2: Torques	1	0 ... 7

The SENSO-Stick responds with the values depending on the chosen data set:

<i>System parameters</i>				
Identifier: 1B $\diamond_h$				
DLC: 8 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Data set		1...2	1, 2	0 ... 15
Value 1			3, 4	0 ... 15
Value 2			5, 6	0 ... 15
Value 3			7, 8	0 ... 15

with the appropriate values:

<i>Byte 1</i>	<i>Byte 2</i>	<i>Byte 3, 4</i>	<i>Byte 5, 6</i>	<i>Byte 7, 8</i>
<i>Data set</i>	-	<i>Parameter 1</i>	<i>Parameter 2</i>	<i>Parameter 3</i>
01 _h	-	Position increments / revolution ¹ Unit: [inc/rev]	-	-
02 _h	-	Nominal Torque ¹ Unit: [mNm]	Maximum Torque ¹ Unit: [mNm]	-

¹ on Motor side

7.4 Read-Only Values

With the following CAN message several read-only values can be read.

Data Request for read-only values				
Identifier: 27 $\diamond_h$				
DLC: 1 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Data set	-	1: Temperatures 2: Av. Motor Torque 3: Error codes	1	0 ... 7

The SENSO-Stick responds with the current values on the chosen data set:

Response with read-only values				
Identifier: 17 $\diamond_h$				
DLC: 8 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Data set		1, 2, 5, 6	1, 2	0 ... 15
Value 1			3, 4	0 ... 15
Value 2			5, 6	0 ... 15
Value 3			7, 8	0 ... 15

with the appropriate values:

Byte 1	Byte 2	Byte 3, 4	Byte 5, 6	Byte 7, 8
Data set	-	Value 1	Value 2	Value 3
01 $_h$	-	Temperature Power Stage Unit: [°C]	Temperature Processor Unit: [°C]	Temperature Chopper Unit: [°C]
02 $_h$	-	Average of Absolute Motor Torque Unit: [mNm]	-	-
03 $_h$	-	Error	Collective Error	-

Error codes of collective errors of the SENSO-Stick are as follows:

<i>Value</i>	<i>Description</i>
00 _h	No Error
01 _h	SENSO-Stick not referenced

7.5 Version

To read the version and variants of the hardware and firmware the following CAN message is provided:

Request Version Information				
Identifier: 28 $\diamond_h$				
DLC: 1 Byte				
Direction: Host → SENSO-Stick				
	Unit	Range	Byte	Bits
Data set	-	1: Firmware version 2: Product version	1	0 ... 7

The SENSO-Stick responds with the appropriate values:

Version Information				
Identifier: 18 $\diamond_h$				
DLC: 8 Byte				
Direction: SENSO-Stick → Host				
	Unit	Range	Byte	Bits
Data set		1...2	1, 2	0 ... 15
Value 1			3, 4	0 ... 15
Value 2			5, 6	0 ... 15
Value 3			7, 8	0 ... 15

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Data set	-	Value 1		Value 2		Value 3	
1	-	Firmware Basic Version		Project Code		Project Version	
		Minor ver.	Major ver.			Minor ver.	Major ver.
2	-	Product Code		Hardware Option			

The firmware of the SENSO-Stick is the basic version. The project code of the SENSO-Stick is by default 3, the project version is the specially adapted firmware variant. The project version "0.0" is the standard version.

Values of Product Code:

	Product Code
SENSO-Stick	200

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